

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
Dot .Net Core and Progressive Web Application Practical
Practical – VII

Roll No.: T052	Name: Divyanjali yadav
Class: TYIT	Batch:01
Date of Assignment: 04-08-2026	Date/Time of Submission: 04-08-2026

- a. Develop a PWA that works in offline mode using Service Worker.

Code:

manifest.json

```
{
  "name": "HTML TO PWA",
  "short_name": "HTP",
  "start_url": "/",
  "display": "standalone",
  "icons": [
    {
      "src": "icon/icon.png",
      "sizes": "512X512",
      "type": "image/png",
      "purpose": "any"
    }
  ]
}
```

service-worker.js

```
self.addEventListener("install", e => {
  console.log("Service workers Installed");
});
self.addEventListener("fetch", e => {
  e.responseWith(fetch(e.request));
});
```

index.html

```
<!DOCTYPE html>
<html>
<head>
  <meta charset="utf-8" />
  <title></title>
</head>
<body>
  <h2>offline</h2>
</body>
</html>
```

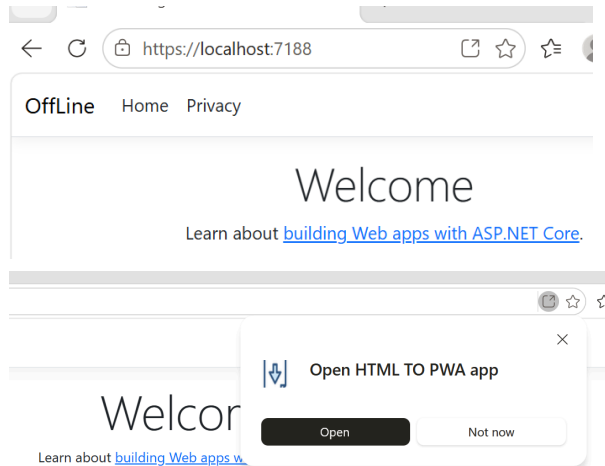
#Add to _Layout.cshtml

#Inside the <head> section:

```
<link rel="manifest" href="/manifest.json"/>
```

#Before the closing </body> tag:

```
<script>
  if('serviceWorker' in navigator){
    navigator.serviceWorker.register('/service-worker.js');
  }
</script>
```

Output:-**b. Create a Student Record Management PWA.****Code:**

```
<div class="text-center">
  <h1 class="display-4">Welcome</h1>
  <p>Learn about <a href="https://learn.microsoft.com/aspnet/core">building Web apps with ASP.NET
Core</a>.</p>
  <h1>Offline Page</h1>
  <h2>Student Record</h2>
```

```
<p>Roll No : @Model.RollNo</p>
```

```
<p>Name : @Model.Name</p>
```

```
</div>
```

Controller:

```
public IActionResult Index()
```

```
{
  Student s = new Student();
```

```
  s.RollNo = 101;
```

```
  s.Name = "Rahul";
```

```
  return View(s);
```

```
}
```

Student.cs

```
namespace OffLine.Models
```

```
{
```

```
  public class Student
```

```
  {
```

```
    public int RollNo { get; set; }
```

```

        public string Name { get; set; }
    }
}
service-worker.js
const CACHE = "v1";

self.addEventListener("install", e => {
    e.waitUntil(
        caches.open(CACHE).then(cache => {
            return cache.addAll([
                "/",
                "/Home/Index",
                "/css/site.css",
                "/js/site.js"
            ]);
        })
    );
});

self.addEventListener("fetch", e => {
    e.respondWith(
        caches.match(e.request).then(response => {
            return response || fetch(e.request);
        })
    );
});

```

Output:

